

Density transformation in R^n

Theorem 7.5.3 Change of Variables in R^n @Judd1998

Assume that $\Phi : X \subset R^n \rightarrow R^n$ is C^1 and that there is a C^1 inverse map $\Phi^{-1} : Y \subset R^n \rightarrow X$ where $Y = \Phi(X)$ and Y and X are open. If f is integrable on X , then

$$\int_Y f(y)dy = \int_X f(\Phi(x))|\det J(\Phi)(x)|dx,$$

where $J(\Phi)(x)$ is the Jacobian of Φ at x .

Corollary: Density transformation

A direct corollary from the change of variables theorem is the density transformation [Section 2.7 in @Hogg2019].

Let f_X be the joint pdf of X , and B any region in the support of Y , and let $A = \Phi^{-1}(B)$. Then, $P(X \in A) = P(\Phi(X) \in \Phi(A)) = P(Y \in B)$. Then, applying Theorem 7.5.3 (@Judd1998),

$$\begin{aligned} P(X \in A) &= \int_A f_X(x)dx \\ &= \int_{\Phi(A)} f_X(\Phi^{-1}(y))|\det J\Phi^{-1}(y)|dy \\ &= \int_B f_X(\Phi^{-1}(y))|\det J\Phi^{-1}(y)|dy \end{aligned}$$

Since B is arbitrary and using $J\Phi^{-1}(y) = [J\Phi(x)]^{-1}$ (inverse function theorem) at $x = \Phi^{-1}(y)$, then the joint pdf of Y must be,

$$f_Y(y) = \frac{f_X(x)}{|\det J\Phi(x)|} \Big|_{x=\Phi^{-1}(y)} \quad (1)$$

Equation 1 is what is applied to my setting.

Applying it to my model

We want the density of observables $f_Y = f_{\mathcal{V}, \tilde{\mathcal{W}}}(\mathcal{V}, \tilde{\mathcal{W}})$ transforming the density of the primitives $f_X = f_{\psi, \omega, \varepsilon}(\psi, \omega, \varepsilon)$.

Because of independence $f_{\psi, \omega, \varepsilon}(\psi, \omega, \varepsilon) = f_{\psi}(\psi) f_{\omega}(\omega) f_{\varepsilon}(\varepsilon)$.

Define $G(\psi, \omega) \equiv u(e^*(\psi, \omega))$, where e^* inverts the FOC $\psi = h(e) - \delta_0 + \delta_1 \omega - \delta_2 \omega^2$ and $u = \ln \frac{M_{it} + e}{M_{it}}$, with M_{it} the true (unobserved) materials level — fixed as e varies, since it comes from a separate production-side FOC that doesn't involve e ($M \perp e \mid \omega$). Note $M_{it}^* = M_{it} + e$ is *not* the right thing to hold fixed here: it's an outcome that moves whenever e does.

Keep the auxiliary $a = \varepsilon$ and let $\Phi : (\psi, \omega, \varepsilon) \mapsto (\mathcal{V}, \tilde{\mathcal{W}}, a)$ to match the theorem's dimension requirement ($\Phi : R^3 \rightarrow R^3$). We will integrate out the auxiliary afterwards.

$$\mathcal{V} = G(\psi, \omega) - \varepsilon, \quad \tilde{\mathcal{W}} = \omega + (1 - \beta)\varepsilon, \quad a = \varepsilon$$

Having defined the transformation Φ , we can get the Jacobian,

$$J\Phi = \frac{\partial(\mathcal{V}, \tilde{\mathcal{W}}, a)}{\partial(\psi, \omega, \varepsilon)} = \begin{pmatrix} G_{\psi} & G_{\omega} & -1 \\ 0 & 1 & 1 - \beta \\ 0 & 0 & 1 \end{pmatrix}$$

read directly off the three equations above ($G_{\psi} \equiv \partial G / \partial \psi$, $G_{\omega} \equiv \partial G / \partial \omega$).

Expand along column 1 (only G_{ψ} is nonzero there):

$$\det J\Phi = G_{\psi} \cdot \det \begin{pmatrix} 1 & 1 - \beta \\ 0 & 1 \end{pmatrix} = G_{\psi} \cdot 1 = G_{\psi}$$

G_{ω} never enters — every cofactor multiplying it comes from a zero in column 1. Mechanically: $\tilde{\mathcal{W}} = \omega + (1 - \beta)\varepsilon$ has **unit coefficient on** ω , a flat translation, so it contributes no curvature. A Jacobian only appears where the map is genuinely curved, the $\psi \rightarrow \mathcal{V}$ link through G , not $\omega \rightarrow \tilde{\mathcal{W}}$.

Since $G = u \circ e^*$ is a composition, by the chain rule $G_{\psi} = u'(e^*) \cdot e_{\psi}^*$.

From $u(e) = \ln(M_{it} + e) - \ln M_{it}$, with M_{it} fixed,

$$u'(e) = \frac{1}{M_{it} + e}$$

Evaluated at $e = e^*$, the accounting identity $M_{it} + e^* = M_{it}^*$ gives $u'(e^*) = 1/M_{it}^*$ directly in terms of the **observed** total — no need to pass through the unobserved M_{it} at all.

Where the earlier draft went wrong. The previous version of this note wrote $u(e) = \ln M_{it}^* - \ln(M_{it}^* - e)$ and differentiated holding M_{it}^* fixed, giving $u'(e) = 1/(M_{it}^* - e) = 1/M_{it}$. That's a partial-vs-total-derivative error: the chain-rule step $G_\psi = u'(e^*) \cdot e_\psi^*$ is only valid if $u(\cdot)$'s *other* argument is genuinely constant as ψ (hence e^*) varies with ω fixed. $M_{it}^* = M_{it} + e^*(\psi, \omega)$ fails that — it moves whenever e^* does — so treating it as fixed silently assumes $\partial M_{it}^* / \partial e = 0$, which is false. M_{it} is the quantity that's actually invariant to ψ here (materials FOC doesn't involve e), so it's the one that belongs inside a single-variable $u(e)$. Confirmed by redoing the calculation as a full total derivative of $G(\psi, \omega) = \ln M^*(\psi, \omega) - \ln(M^*(\psi, \omega) - e^*(\psi, \omega))$, correctly letting M^* move with e^* : it collapses to the same e_ψ^* / M_{it}^* . The *value* of u_{it} is identical either way (both are $\ln(M_{it}^* / M_{it})$) — only the derivative used in the Jacobian differs, by a factor of M_{it}^* / M_{it} .

To obtain e_ψ^* , differentiate the FOC identity $\psi = h(e^*(\psi, \omega)) - \delta_0 + \delta_1 \omega - \delta_2 \omega^2$ w.r.t. ψ (implicit function theorem, valid since $h' \neq 0$):

$$1 = h'(e^*) \cdot e_\psi^* \implies e_\psi^* = \frac{1}{h'(e^*)}$$

Combining both,

$$G_\psi = \frac{1}{M_{it}^* h'(e)}$$

From the Equation 1, the multiplying factor is $1/|\det J\Phi| = 1/|G_\psi|$:

$$J_i(\varepsilon) \equiv \frac{1}{|G_\psi|} = |h'(e)| M_{it}^*$$

Every piece is traceable: $|h'(e)|$ from inverting the FOC ($\psi \rightarrow e$), M_{it}^* from the log-ratio definition ($e \rightarrow \mathcal{V}$, holding the true M_{it} fixed). They multiply because $G = u \circ e^*$ is a composition, so the chain rule stacks them.

Applying the density transformation formula gives the joint density of $(\mathcal{V}, \tilde{\mathcal{W}}, a)$. However, only $(\mathcal{V}, \tilde{\mathcal{W}})$ are observed; a was bookkeeping.

Integrating it a out (an ordinary single integral, not part of the theorem) gives the likelihood contribution of firm i :

$$f_{\mathcal{V}, \tilde{\mathcal{W}}}(\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}) = \int f_\psi(\psi(\varepsilon)) J_i(\varepsilon) f_\omega(\omega(\varepsilon)) f_\varepsilon(\varepsilon) d\varepsilon \quad (2)$$

(f_ω here is left generic — *which* density of ω belongs in this slot, and what it's implicitly conditioned on, is exactly what “Extending to a dynamic likelihood” below has to pin down; don't read the bare unconditional marginal into it yet.)

A 3-for-3 change of variables that ends in a 1-D integral: the theorem handles the square transformation and hands back a Jacobian; marginalizing the unobserved auxiliary coordinate is the separate, ordinary step left over once the theorem's job is done.

Sanity check: is M_{it}^* missing from the likelihood?

My supervisor’s question, roughly: we observe M_{it}^* (the overreported materials figure) directly in the data — shouldn’t the likelihood be the *joint* density $f_{\mathcal{V}, \tilde{\mathcal{W}}, M^*}$, or at least $f_{\mathcal{V}, \tilde{\mathcal{W}}|M^*}$, rather than the unconditional $f_{\mathcal{V}, \tilde{\mathcal{W}}}$ used in Equation 2? This section works out the algebra to answer that precisely — and the answer is: neither. M_{it}^* is already fully inside $\mathcal{V}_{it}$; it isn’t missing, and it shouldn’t be conditioned on either.

What $\mathcal{V}$ and $\tilde{\mathcal{W}}$ actually contain

From `56-id-evasion.qmd` (lowercase = logs; k, l, y, m^* are the raw observables, $c_1 \equiv \ln \rho_t - \ln P_t - \ln \beta - \ln \mathcal{E}$ a constant already pinned down in stage 1):

$$\mathcal{V}_{it} = m_{it}^* - y_{it} + c_1 \quad (\text{eq-ob-ev, rearranged})$$

$$\tilde{\mathcal{W}}_{it} = y_{it} - \beta m_{it}^* + \beta \mathcal{V}_{it} - \alpha_K k_{it} - \alpha_L l_{it} \quad (\text{eq-big-w} + \text{eq-w-squiggle})$$

Substitute the first equation into the second — the m_{it}^* terms cancel exactly:

$$\tilde{\mathcal{W}}_{it} = (1 - \beta) y_{it} + \beta c_1 - \alpha_K k_{it} - \alpha_L l_{it} \quad (3)$$

That cancellation is not an accident. The text right after `eq-big-w` in `56-id-evasion.qmd` says it in words: $\tilde{\mathcal{W}}$ is built by “trading unobservables, the unobserved tax evasion for the output shock” — i.e. $\tilde{\mathcal{W}}$ is *constructed* to purge tax evasion from the productivity residual. Once that trade is made, of course $\tilde{\mathcal{W}}$ can’t also be carrying information about m_{it}^* — that information has been swept entirely into $\mathcal{V}_{it}$.

$(\mathcal{V}, \tilde{\mathcal{W}}) \leftrightarrow (y, m^*)$ is a linear bijection

Given (k_{it}, l_{it}) , treat (y_{it}, m_{it}^*) as the two “new” pieces of randomness for firm i at t (everything else — $k, l, \rho_t, P_t, \hat{\beta}, \mathcal{E}$ — is either predetermined or already estimated in stage 1). The pair of equations above is a linear (affine) map

Measurement caveat. m_{it}^* and y_{it} here denote the deflated real-log quantities the *production* identity needs, not primitives independently observed the same way $\mathcal{V}_{it}$ is. $\mathcal{V}_{it}$ itself is built directly from the nominal materials-to-sales ratio (`eq-eva-emp` in `600-opt-tax.qmd`) — ρ_t, P_t cancel in that ratio, so $\mathcal{V}_{it}$ never required m_{it}^* or y_{it} to be separately deflated in the first place. The bijection argument below is unaffected (it only needs $\det A \neq 0$), but treat m_{it}^*, y_{it} as constructed objects here,

not as separately-clean survey data. See `Paper/sections/9999-elvis.qmd`'s “note on measurement” and the corresponding `CLAUDE.md` entry.

$$\Phi : (y_{it}, m_{it}^*) \mapsto (\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}), \quad \begin{pmatrix} \mathcal{V} \\ \tilde{\mathcal{W}} \end{pmatrix} = \underbrace{\begin{pmatrix} -1 & 1 \\ 1 - \beta & 0 \end{pmatrix}}_A \begin{pmatrix} y \\ m^* \end{pmatrix} + \begin{pmatrix} c_1 \\ \beta c_1 - \alpha_K k - \alpha_L l \end{pmatrix}.$$

Reminder: linear (affine) bijection. A map $\Phi(x) = Ax + b$ on $\mathbb{R}^n$ is a bijection **iff** $\det A \neq 0$; when it is, the inverse is affine too, $\Phi^{-1}(y) = A^{-1}(y - b)$, and — this is the part worth remembering — its Jacobian is A^{-1} , a **constant matrix**. It doesn't vary point to point the way the Jacobian of a curved map does. That's a much stronger statement than “invertible at this particular point,” which is all Theorem 7.5.3 promises for a general C^1 map. It's *why* checking $\det A \neq 0$ once is enough here, instead of needing an implicit-function-theorem argument at each (y, m^*) .

Here $\det A = (-1)(0) - (1)(1 - \beta) = -(1 - \beta)$, and $0 < \beta < 1$ always (materials share), so $\det A \neq 0$ unconditionally — Φ is a bijection for every firm, every θ , no knife-edge case to worry about.

That bijection is the whole answer to the supervisor's question. It means $(\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it})$ and (y_{it}, m_{it}^*) are two labels for the exact same information — nothing is lost by working in $(\mathcal{V}, \tilde{\mathcal{W}})$ -space, and nothing is gained by tacking M_{it}^* back on as though it were a separate fact. M_{it}^* is already inside $\mathcal{V}_{it}$, one-for-one. The three cases below spell out why that rules out (b) and (c); a separate aside afterward works out the actual size of Φ 's Jacobian, which is a related but different question from the one just answered.

Three candidate objects

(a) $f_{\mathcal{V}, \tilde{\mathcal{W}}|K, L}$ — **the right target, but not simply Equation 2 as originally written.** Because Φ is a bijection, $f_{\mathcal{V}, \tilde{\mathcal{W}}|K, L}$ is just the joint density of the two raw endogenous outcomes (Y_{it}, M_{it}^*) conditional on the predetermined state (K_{it}, L_{it}) , reparametrized through Φ^{-1} and rescaled by Φ 's Jacobian (a θ -free constant — worked out in the aside below). Nothing about M_{it}^* 's randomness is missing; it's just been relabeled — that part of the argument is unaffected by anything below.

Correction. An earlier version of this note stopped here and declared this equal to Equation 2 as originally written (bare, unconditional f_ω). That equality only holds under the extra assumption $\omega_{it} \perp (K_{it}, L_{it})$ — which turns out to be untenable: ω is AR(1)-persistent and K is a dynamically accumulated state chosen anticipating that persistence, and asserting independence here would strip the very AR(1) moment conditions that identify α_K, α_L of any instrument relevance. The correct formula

replaces bare f_ω with the AR(1) innovation density f_η , conditioned on the firm's own lagged $\tilde{\mathcal{W}}_{it-1}$ (a sufficient statistic for (K, L) 's relevance to ω_{it}) — worked out in “Extending to a dynamic likelihood” below, Equation 5.

(b) $f_{\mathcal{V}, \tilde{\mathcal{W}}, M^*}$ — ill-posed, don't use. Given $(\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}, K_{it}, L_{it})$, M_{it}^* is *not* an independent third random draw — it's pinned down exactly by rearranging the definition of $\mathcal{V}_{it}$ itself (eq-ob-ev above): $m_{it}^* = \mathcal{V}_{it} + y_{it} - c_1$. The joint law of $(\mathcal{V}, \tilde{\mathcal{W}}, M^*)$ given (K, L) lives entirely on a 2-dimensional surface inside $\mathbb{R}^3$ (the graph of Φ^{-1}), which has zero Lebesgue measure in $\mathbb{R}^3$. A genuine 3-D density doesn't exist for it — you'd need a Dirac delta pinning M^* to that surface, and multiplying that in just reproduces (a) with an extra factor of “1” dressed up as a third dimension. Intuitively: there are only two independent sources of firm- it randomness feeding this part of the likelihood (Y_{it} and M_{it}^* , equivalently ε_{it} and u_{it} 's realization); asking for a *third* independent coordinate over-counts.

Counterexample: try to add M^* , watch the extended matrix degenerate.

M_{it}^* isn't a new draw — it's just m_{it}^* itself. Tack it on as a third output of the same linear map:

$$\begin{pmatrix} \mathcal{V} \\ \tilde{\mathcal{W}} \\ M^* \end{pmatrix} = \underbrace{\begin{pmatrix} -1 & 1 \\ 1-\beta & 0 \\ 0 & 1 \end{pmatrix}}_{A_{\text{ext}}} \begin{pmatrix} y \\ m^* \end{pmatrix} + \begin{pmatrix} c_1 \\ \beta c_1 - \alpha_K k - \alpha_L l \\ 0 \end{pmatrix}.$$

A_{ext} is 3×2 : three outputs built from only two free primitives (y, m^*) . Three vectors in $\mathbb{R}^2$ (its three rows) can never be linearly independent — $\text{rank}(A_{\text{ext}}) \leq 2 < 3$ on dimension-counting grounds alone, no computation needed. Eliminating (y, m^*) between the three defining equations pins down the exact dependency: taking $(1-\beta) \times (\text{row } 1) + 1 \times (\text{row } 2) - (1-\beta) \times (\text{row } 3)$ cancels y and m^* entirely, leaving

$$(1-\beta)M_{it}^* - (1-\beta)\mathcal{V}_{it} - \tilde{\mathcal{W}}_{it} = \alpha_K k_{it} + \alpha_L l_{it} - c_1.$$

This is a *fixed, non-random* affine constraint linking $(\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}, M_{it}^*)$ given (k_{it}, l_{it}) — the algebraic fingerprint of “lives on a 2-D surface in $\mathbb{R}^3$ ”: not a description, an exact equation. And if you insist on making the map square, as Theorem 7.5.3 literally requires ($\Phi : \mathbb{R}^n \rightarrow \mathbb{R}^n$), the only way is to invent a third primitive z for M^* to depend on — but none exists in the model. Appending one that (honestly) enters none of the three equations gives

$$J\Phi_{\text{ext}} = \frac{\partial(\mathcal{V}, \tilde{\mathcal{W}}, M^*)}{\partial(y, m^*, z)} = \begin{pmatrix} -1 & 1 & 0 \\ 1-\beta & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \det J\Phi_{\text{ext}} = 0$$

a zero column forces $\det = 0$ outright. Theorem 7.5.3's invertibility hypothesis fails by construction — (b) isn't just a bad modeling choice, it's algebraically ill-posed.

(c) $f_{\mathcal{V}, \tilde{\mathcal{W}}|M^*}$ — **well-defined, but the wrong target.** This one is a legitimate density (conditioning on one coordinate of a bijection just fixes it), but conditioning on M_{it}^* throws away exactly the randomness the model is supposed to explain. $M_{it}^* = M_{it} + e_{it}^*(\psi_{it}, \omega_{it}; \theta)$ is the firm’s *endogenous* evasion outcome — the object generated by the evasion FOC — and it’s the channel through which the benefit shifter $\tau\rho_t$ moves evasion and thereby identifies λ (see the “Identification of λ/k ” note in the model doc). Treating M_{it}^* as given and conditioning on it is like conditioning a wage regression on the wage: what’s left to fit is just $\tilde{\mathcal{W}}_{it}$, i.e. y_{it} alone — a plain productivity-process likelihood with no remaining leverage on θ at all.

So: M_{it}^* is already fully determined, given $(\mathcal{V}_{it}, Y_{it})$, by the bijection — it isn’t missing from Equation 2, and there’s nothing to gain by adding it back in. (b) asks for a coordinate that doesn’t independently exist; (c) throws away the randomness the model exists to explain; (a) already has M_{it}^* , for free, exactly because Φ is invertible.

The Jacobian of Φ , and why it’s not $J_i(\varepsilon)$

The bijection argument above only needed $\det A \neq 0$ — it never needed the actual *value* of the Jacobian. A separate, narrower question: translating f_{y, m^*} into $f_{\mathcal{V}, \tilde{\mathcal{W}}}$ does pick up a Jacobian factor from the density-transformation formula (Equation 1) — could that factor be hiding some θ -dependence that Equation 2 isn’t accounting for?

Solving Φ^{-1} :

$$y_{it} = \frac{\tilde{\mathcal{W}}_{it} + \alpha_K k_{it} + \alpha_L l_{it} - \beta c_1}{1 - \beta}, \quad m_{it}^* = \mathcal{V}_{it} + y_{it} - c_1,$$

with constant Jacobian $|\det A^{-1}| = 1/(1 - \beta)$ — a number that depends only on $\hat{\beta}$ (fixed, from stage 1), never on $\theta = (\lambda, \delta_0, \delta_1, \delta_2)$. That matters: it means relabeling from (y, m^*) -space to $(\mathcal{V}, \tilde{\mathcal{W}})$ -space can’t itself be doing any estimation work. A constant can’t move the argmax of the likelihood over θ — so no, nothing is hiding in it.

Contrast that with $J_i(\varepsilon) = |h'(e)| M_{it}^*$ (derived earlier in this note), which comes from a different map entirely, $(\psi, \omega, \varepsilon) \rightarrow (\mathcal{V}, \tilde{\mathcal{W}}, a)$ — the one that actually appears inside Equation 2. That map is **nonlinear** ($G = u \circ e^*$ curves), so its Jacobian varies pointwise with e (hence with ε), is only *locally* invertible, and depends on θ through λ (inside h') — so it *is* doing estimation work and can’t be dropped. The two Jacobians both happen to involve M_{it}^* , which invites confusion, but they answer different questions: this section’s Jacobian just confirms the $(\mathcal{V}, \tilde{\mathcal{W}})$ reparametrization is free — it’s never actually used in estimation, since the likelihood is built directly in $(\mathcal{V}, \tilde{\mathcal{W}})$ -space from the primitives; $J_i(\varepsilon)$ is a load-bearing piece of the likelihood itself.

Summary

- M_{it}^* is in the likelihood — one-for-one, via $\mathcal{V}_{it}$, given (K_{it}, L_{it}) , because Φ is a bijection.
- The right object is $f_{\mathcal{V}, \tilde{\mathcal{W}}|K, L}$ — but its correct formula is Equation 5 (see “Extending to a dynamic likelihood” below), not the bare- f_ω Equation 2 as first written: that required the extra, rejected assumption $\omega \perp (K, L)$.
- $f_{\mathcal{V}, \tilde{\mathcal{W}}, M^*}$ double-counts — no genuine third dimension of randomness exists.
- $f_{\mathcal{V}, \tilde{\mathcal{W}}|M^*}$ conditions away the model’s own endogenous outcome and would leave θ unidentified.
- Φ ’s own Jacobian, $1/(1 - \beta)$, is a θ -free constant — it’s not secretly doing estimation work, unlike $J_i(\varepsilon)$ inside Equation 2.

Extending to a dynamic likelihood: conditioning on ω_{it-1} instead of (K, L)

The sanity check above correctly identified $f_{\mathcal{V}, \tilde{\mathcal{W}}|K, L}$ as the right target, but flagged its own overclaim that this equals Equation 2 as originally written. This section works out the actual formula. Unlike M_{it}^* (redundant by a *bijection*, an algebraic fact true regardless of the DGP), whether conditioning on (K, L) matters for ω is a *distributional* question — and it does matter here, because ω is AR(1)-persistent while K is a dynamically accumulated state chosen anticipating exactly that persistence. Asserting $\omega_{it} \perp (K_{it}, L_{it})$ to make the issue disappear would be inconsistent with the very AR(1) moment conditions (**eq-eta-moments** in **56-id-evasion.qmd**) that identify α_K, α_L in the first place — if ω_{it} carried no relationship to (K_{it}, L_{it}) , there would be no instrument-relevance left for that regression to exploit.

Notation clash, resolved locally. **56-id-evasion.qmd**’s AR(1) drift (**eq-prod-markov**) is called h , with linear case $h(\xi) = \delta_0 + \delta_1 \xi$ (**eq-big-w-reg**) — both symbols are already spoken for in this note (h is the cost-FOC inverse map from Equation 2; $\delta_0, \delta_1, \delta_2$ are κ ’s U-shape parameters). Purely so formulas below stay parseable, rename the AR(1) drift here only: $g(\xi) = \gamma_0 + \gamma_1 \xi$, η_{it} kept as in the source paper. **56-id-evasion.qmd** itself is untouched/read-only — translate when cross-referencing.

Why (K, L) -conditioning reduces to ω_{it-1} -conditioning

$\omega_{it} = g(\omega_{it-1}) + \eta_{it}$. (K_{it}, L_{it}) are functions of the firm’s $t - 1$ information set (predetermined); η_{it} is news arriving strictly after that, independent of $(\omega_{it-1}, K_{it}, L_{it})$ jointly — exactly what the paper’s orthogonality table already asserts for η_{it} (orthogonal to k_{it}, l_{it} **and** to ω_{it-1}). Conditional on ω_{it-1} , the only randomness left in ω_{it} is η_{it} , which doesn’t depend on (K_{it}, L_{it}) :

$$f_{\omega_{it}|\omega_{it-1}, K_{it}, L_{it}}(\omega_{it}) = f_\eta(\omega_{it} - g(\omega_{it-1})) = f_{\omega_{it}|\omega_{it-1}}(\omega_{it}).$$

So ω_{it-1} is a **sufficient statistic**: once it's known, (K_{it}, L_{it}) adds nothing (extending this to the full triple $(\psi_{it}, \omega_{it}, \varepsilon_{it})$ needs $\psi_{it}, \varepsilon_{it} \perp (K_{it}, L_{it}) \mid \omega_{it-1}$ too, already on the same footing as the model's other idiosyncratic-shock assumptions). This licenses building the likelihood directly around ω_{it-1} — the true state variable — rather than around (K, L) , which was only ever a noisy proxy for it and, as flagged in the sanity check, leads to a regress $(f_{\omega_{it-1}|K,L})$ is exactly as hard a problem as the one just solved, one period earlier) if pursued directly.

Step 1 — ω_{it} 's density conditional on the lag

$$f_{\omega_{it}|\omega_{it-1}}(\omega_{it}) = f_{\eta}(\omega_{it} - g(\omega_{it-1})).$$

Step 2 — redo the change of variables, ω_{it-1} held fixed

The Φ map is untouched — $\mathcal{V}_{it} = G(\psi_{it}, \omega_{it}) - \varepsilon_{it}$, $\tilde{\mathcal{W}}_{it} = \omega_{it} + (1 - \beta)\varepsilon_{it}$, $a = \varepsilon_{it}$ — and so is its Jacobian $J_i(\varepsilon_{it}) = |h'(e)| M_{it}^*$: ω_{it-1} never enters the map, exactly like (K, L) before it, so it only ever shifts *which* ω -density gets plugged in, never the curvature. Substituting the conditional density from Step 1 for the marginal f_{ω} in Equation 2:

$$f_{\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}|\omega_{it-1}}(\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}) = \int f_{\psi}(\psi(\varepsilon_{it})) J_i(\varepsilon_{it}) f_{\eta}(\omega(\varepsilon_{it}) - g(\omega_{it-1})) f_{\varepsilon}(\varepsilon_{it}) d\varepsilon_{it}. \quad (4)$$

Step 3 — ω_{it-1} isn't observed; peel it off using $\tilde{\mathcal{W}}_{it-1}$

From **eq-w-squiggle**, $\omega_{it-1} = \tilde{\mathcal{W}}_{it-1} - (1 - \beta)\varepsilon_{it-1}$. $\tilde{\mathcal{W}}_{it-1}$ is firm- i data — already constructed and used to estimate the AR(1) in stage 1 — leaving only ε_{it-1} unknown, with known density f_{ε} (same as stage 1, by stationarity) and independent of everything else in Equation 4: $\varepsilon_{it-1} \perp \varepsilon_{it}, \eta_{it}$ is in the orthogonality table directly; $\varepsilon_{it-1} \perp \psi_{it}$ is not — it's a new maintained assumption introduced here, extending the model's existing $\psi \perp \varepsilon$ (same period) across time, on the same footing as that assumption rather than independently established. Marginalizing ε_{it-1} out is therefore an ordinary integral — no Jacobian, same bookkeeping move already used to dispose of $a = \varepsilon_{it}$ in the original derivation, not a new instance of the theorem:

$$f_{\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}|\tilde{\mathcal{W}}_{it-1}}(\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}) = \iint f_{\psi}(\psi(\varepsilon_{it})) J_i(\varepsilon_{it}) f_{\eta}(\omega(\varepsilon_{it}) - g(\tilde{\mathcal{W}}_{it-1} - (1 - \beta)\varepsilon_{it-1})) f_{\varepsilon}(\varepsilon_{it}) f_{\varepsilon}(\varepsilon_{it-1}) d\varepsilon_{it} d\varepsilon_{it-1}. \quad (5)$$

This is the corrected likelihood contribution: Equation 2 with the unconditional f_{ω} replaced by the AR(1) innovation density f_{η} , evaluated at a lag-productivity argument that is itself integrated against the one genuinely unknown piece, ε_{it-1} .

Recovering f_η

Worked out via `eq-big-w-reg` (renamed): define the fully constructible residual

$$\zeta_{it} \equiv \mathcal{W}_{it} - \alpha_K k_{it} - \alpha_L l_{it} - \gamma_0 - \gamma_1 (\mathcal{W}_{it-1} - \alpha_K k_{it-1} - \alpha_L l_{it-1}) = (1-\beta)\varepsilon_{it} - \gamma_1(1-\beta)\varepsilon_{it-1} + \eta_{it},$$

a sum of three mutually independent pieces (orthogonality table again), the first two known up to the already-deconvolved stage-1 f_ε . Fourier deconvolution recovers the third:

$$\phi_\eta(t) = \frac{\phi_\zeta(t)}{\phi_\varepsilon((1-\beta)t) \phi_\varepsilon(-\gamma_1(1-\beta)t)} \rightarrow f_\eta,$$

the same deconvolution class as stage 1's f_e/f_ε recovery, just against a two-factor known-error density instead of one.

Consistency check: nests the static case

If productivity had no persistence, $g(\xi) \equiv \gamma_0$ (constant, $\gamma_1 = 0$), the ε_{it-1} integral in Equation 5 becomes a no-op — $f_\eta(\omega(\varepsilon_{it}) - \gamma_0)$ doesn't depend on ε_{it-1} , so it integrates out of a term that doesn't contain it, leaving exactly Equation 2 with $f_\omega(\cdot) \equiv f_\eta(\cdot - \gamma_0)$. The dynamic likelihood collapses to the static one exactly when the thing that made (K, L) -conditioning necessary — persistence — is switched off. That's the intended sanity check: the fix activates precisely in proportion to γ_1 , not unconditionally.

Practical caveat (not resolved here): Equation 5 needs $\tilde{\mathcal{W}}_{it-1}$, so it only applies to firm-periods with an observed predecessor. A firm's first observed period has no lag to condition on — the standard dynamic-panel initial-conditions problem (e.g. Heckman 1981); such periods would fall back to Equation 2's unconditional f_ω (interpreted as ω 's stationary/ergodic marginal) rather than Equation 5. Flagging this rather than resolving it — it's a separate design choice from the derivation above.